

Practical 12

```
In [2]: import numpy as np
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import AgglomerativeClustering
from sklearn.metrics import silhouette_score
import matplotlib.pyplot as plt
import scipy.cluster.hierarchy as sch

# 1. Load the Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# 2. Preprocess the data: Scale the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 3. Apply Hierarchical Clustering (Agglomerative Clustering)
agg_clustering = AgglomerativeClustering(n_clusters=3, metric='euclidean', linkage='ward')
y_pred = agg_clustering.fit_predict(X_scaled)

# 4. Evaluate the model using Silhouette Score
sil_score = silhouette_score(X_scaled, y_pred)
```

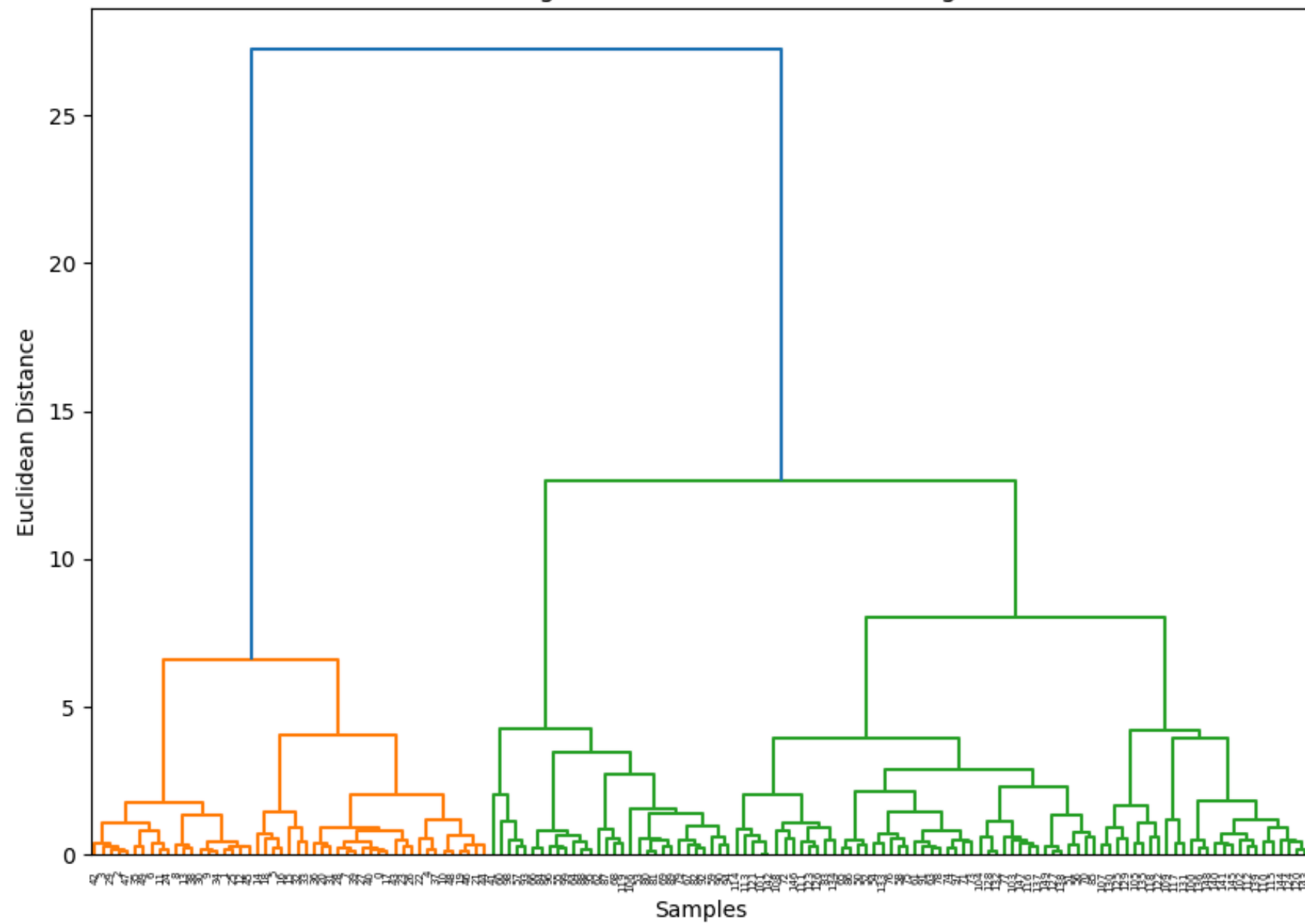
```
# 5. Print the evaluation metrics
print(f"Silhouette Score: {sil_score}")

# 6. Visualize the Dendrogram to understand the hierarchical structure
plt.figure(figsize=(10, 7))
dendrogram = sch.dendrogram(sch.linkage(X_scaled, method='ward'))
plt.title('Dendrogram for Hierarchical Clustering')
plt.xlabel('Samples')
plt.ylabel('Euclidean Distance')
plt.show()
print("\n")

# 7. Visualize the clustering results (2D projection for simplicity)
plt.figure(figsize=(8, 6))
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=y_pred, cmap='viridis', marker='o', s=100)
plt.title('Hierarchical Clustering Results')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.show()
```

Silhouette Score: 0.4466890410285909

Dendrogram for Hierarchical Clustering



Hierarchical Clustering Results

